

Lists

CSCI 1101 – Introduction to Programming in Python

Lists

- In Python, Lists allow you to store multiple literal values in a single variable. For example, if you want to store the names of various grocery items you can use a list.

```
grocery_list = [ "Eggs", "Milk", "Bread" ]
```

Accessing List Elements

- Each element in a list is associated with a number known as an **index**. The index of a list always starts at **0**.
- The index is used as a key to access the element associated to it.

```
languages = [ "Python", "Swift", "C++" ]  
print( languages[ 0 ] ) # will output "Python"
```

List	→	['Python', 'Swift', 'C++']		
Index	→	0	1	2

List	→	['Python', 'Swift', 'C++']		
Access List	→	languages[0]	languages[1]	languages[2]

List Characteristics

- **Ordered:** Elements in a list are ordered by their index.
- **Changeable:** Elements in a list can be changed or removed.
- **Heterogeneous:** Elements in a list can be of different types (including List).
- **Allows Duplicates:** Elements in a list can be duplicates but will have different index.

```
L = [ 20, 'Jessa', 35.75, [30, 60, 90] ]
```

L[0] L[1] L[2] L[3]

List Splicing

- To print out the entire list you can simply use the `print()` method.

```
my_list = [ "dog", "cat", "bird", "fish", "hamster" ]
```

```
print( my_list ) # will output [ "dog", "cat", "bird", "fish", "hamster" ]
```

- To print a portion of the list we can use the splicing operator `:`. Simply indicate the start and ending index.

```
my_list = [ "dog", "cat", "bird", "fish", "hamster" ]
```

```
print( my_list[ 0 : 2 ] ) # will output [ "dog", "cat" ]
```

```
print( my_list[ 2 : 4 ] ) # will output [ "bird", "fish" ]
```

Adding Elements to List

- To add new elements to a list you can use the **append()** method. Appended elements are added to the back of the list.

```
my_list = [ "dog", "cat" ]
```

```
print( my_list ) # will output [ "dog", "cat" ]
```

```
my_list.append("bird" )
```

```
print( my_list ) # will output [ "dog", "cat", "bird" ]
```

Adding Elements to Specific Index in List

- To add new elements to a list but into a specific index, you can use the **insert()** method. Inserted elements are added to the specified index and all other elements are pushed back.

```
my_list = [ "dog", "cat", "bird" ]
```

```
print( my_list ) # will output [ "dog", "cat", "bird" ]
```

```
my_list.insert( 2, "hamster" )
```

```
print( my_list ) # will output [ "dog", "cat", "hamster", "bird" ]
```

Changing List Elements

- To update an element in a list you can simply use the `[]` access operator to specify the index of the element and then use the `=` assignment operator to update the element.

```
my_list = [ "dog", "cat", "bird" ]
```

```
print( my_list ) # will output [ "dog", "cat", "bird" ]
```

```
my_list[ 1 ] = "lizard"
```

```
print( my_list ) # will output [ "dog", "lizard", "bird" ]
```


Removing List Elements

- To remove elements from a list you can use the **del** statement. The del statement can be used to remove a single element, multiple element, or the entire list.

```
my_list = [ "dog", "cat", "bird", "fish", "hamster" ]
```

```
print( my_list ) # will output [ "dog", "cat", "bird", "fish", "hamster" ]
```

```
del my_list[ 1 ]
```

```
print( my_list ) # will output [ "dog", "bird", "fish", "hamster" ]
```

```
del my_list[ 1 : 3 ]
```

```
print( my_list ) # will output [ "dog", "hamster" ]
```

```
del my_list
```

```
print( my_list ) # will output Error, list no longer exists
```

Getting the List Length

- To get the length of a list or the number of elements in a list, you can use the **len()** function.

```
my_list = [ "dog", "cat", "bird" ]  
print( len( my_list ) ) # will output 3
```

Iterating Through a List

- To print the entire list, you can simply use the `print()` function. However, if you wish to iterate through a list to either print the elements individually or access each element individually, you can use a **for** loop.

```
my_list = [ "dog", "cat", "bird" ]
```

```
for item in my_list :
```

```
    print( item )
```

Python List Methods

Method	Description
<code>append()</code>	Adds an item to the end of the list
<code>extend()</code>	Adds items of lists and other iterables to the end of the list
<code>insert()</code>	Inserts an item at the specified index
<code>remove()</code>	Removes the specified value from the list
<code>pop()</code>	Returns and removes item present at the given index
<code>clear()</code>	Removes all items from the list
<code>index()</code>	Returns the index of the first matched item
<code>count()</code>	Returns the count of the specified item in the list
<code>sort()</code>	Sorts the list in ascending/descending order
<code>reverse()</code>	Reverses the item of the list
<code>copy()</code>	Returns the shallow copy of the list